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Homework 3

1.a What Really is a Side-Channel Attack? Mention five commonly exploited side-channel emissions.

A side-channel attack is an attack that extracts secret information by observing physical effects produced while a device is operating. In hardware security, these attacks exploit unintended leakage from the implementation itself. Rather than attacking AES, RSA or another cryptographic algorithm, the attacker studies how the hardware behaves while processing secret data such as keys, plaintext or intermediate values. This is done by observing physical signals such as power consumption, timing behavior and electromagnetic radiation which can reveal sensitive information during operation. Side-channel attacks are most commonly divided into passive attacks, where the attack only observes leakage and active attacks, where the attack also interferes with device operation to increase the leakage or induce faults (Bhunja & Tehranipoor, 2019; Paar & Pelzl, 2014). Listed below are five commonly exploited side-channel emissions:

1. **Power Consumption:** attackers measure the variations in current draw while the chip performs operations
2. **Electromagnetic (EM) radiation:** switching activity inside the device emits electromagnetic signals that can be captured.
3. **Optical emissions:** light emitted or optical changes in the device during operation leak useful information to an attacker

4. **Timing and delay:** the time required to complete an operation can reveal information about secret dependent branches or data paths
5. **Acoustic emissions:** some devices reveal information through sound or vibration produced during processing

This shows that even if a cryptographic algorithm is strong, the physical implementation can still leak enough information to compromise the system (Bhunia & Tehranipoor, 2019; Paar & Pelzl, 2014)

2.a What are Hardware Targets? Mention 2 victims of Hardware Targets.

Devices, components or assets within a computing system are hardware targets that attackers try to compromise through hardware focused attacks. These targets include cryptographic chips, processors, memories, systems-on-chip, printed circuit boards, firmware storage, debug interfaces and other security-sensitive hardware. In hardware security, these targets are especially valuable because they hold or control important assets such as secret keys, firmware, credentials or protected data. Compromising the hardware is dangerous because it can provide low-level access that software protections may not be able to stop (Bhunia & Tehranipoor, 2019). Two common victims of hardware targeting are:

1. **Smart cards:** these are targeted to extract secret keys, bypass authentication or reveal confidential information (Maswood, 2026).
2. **FPGAs:** these are targeted because they are used in security sensitive applications and may be exploited to expose protected data (Maswood, 2026).

These victims are attractive because they store secrets, enforce trust and support critical system functions (Maswood, 2026).

2.b How do clock glitching and voltage glitching techniques work in fault injection attacks?

Clock glitching and voltage glitching are known as fault injection techniques which intentionally disturb the normal electrical operating conditions of a device so that it makes mistakes. Instead of passively observing the leakage like a power or timing attack, the attack will actively create abnormal behavior at carefully chosen moments. The goal is to cause the hardware to either skip instructions, misread data, corrupt memory values or bypass security checks. Fault injection attacks are considered active attacks because they interfere with the way the device behaves during execution (Bhunia & Tehranipoor, 2019). Clock glitching works by introducing abnormal clock pulses or momentarily changing the clock timing. If the chip receives an unexpected pulse, a shortened cycle or an unstable clock transition, this can cause some logic to not settle correctly before the next state is captured. This can cause instruction skipping, incorrect branching, faulty comparisons or corrupted control flow. In secure devices, even a slight clock disturbance may be enough to bypass password checks, authentication routines and memory protection logic. Voltage glitching works by briefly dropping or spiking the supply voltage while the device is executing a critical operation. Since digital circuits depend on a stable voltage level, even a slight disturbance can cause timing violations, bit flips, corrupted reads and incomplete writes (Bhunia & Tehranipoor, 2019; Maswood, 2026).

2.c What are the potential consequences and objectives associated with these techniques in terms of compromising a target system's security and functionality?

The objectives and consequences of these glitching techniques can be severe. An attacker can use them to bypass secure boot, skip authentication checks, corrupt cryptographic calculations, extract secret keys through faulty outputs, unlock debug features and be able to read data from protected memory. In addition to violating confidentiality, these attacks can also affect

integrity by altering computation results and availability by crashing or destabilizing the device (Bhunia & Tehranipoor, 2019).

3.a Explain SPA and DPA along with their characteristics.

Simple Power Analysis (SPA) and Differential Power Analysis (DPA) are two forms of power side-channel attacks that attempt to recover secret information from a cryptographic device by analyzing its power consumption. SPA is the direct interpretation of measuring power traces. The attack visually examines the power waveform of the device while it's performing cryptographic operations and tries to identify patterns that correspond to specific instructions, branches and data-dependent behavior. Its main characteristic is that it can still succeed even with relatively few traces, but it is heavily dependent on clear leakage and attacker expertise (Paar & Pelzl, 2014). DPA is more statistical, rather than relying on obvious visual differences, the attack collects many power traces and uses statistical methods to find correlations between measured power consumption and guesses of secret dependent intermediate values. Its main strength is that it can detect subtle leakage that is not obvious in a single trace, but it usually needs a large number of power measurements for this to succeed. Both attacks are dangerous because power consumption reflects transistor switching activity, which depending on the data being processed, could include secret key material (Paar & Pelzl, 2014).

3.b Explain the Levels of Security and Design for Security.

In hardware security, levels of security are described as the degree and scope of protection applied to the assets in a system. A secure system must protect the confidentiality, integrity and availability of its data and operations. Confidentiality prevents unauthorized disclosure of data, integrity protects against unauthorized modification and availability ensures that only authorized users can access appropriate resources when needed. These core principles

apply across multiple hardware levels, including the chips, boards, embedded systems and full computing platforms (Bhunias & Tehranipoor, 2019).

Design for Security (DfS) means intentionally building security into the hardware design process rather than treating it as an afterthought. This includes incorporating secure architectures, validating information flow, enforcing access-control policies, reducing side-channel leakage, hardening the system against fault injection and checking for malicious modifications. The purpose of DfS is to make the hardware itself resistant to attack throughout the entire design, validation and deployment process. So, the levels of security describe how much and how broadly protection is enforced, while DfS is the strategy of integrating those protections directly into the hardware architecture, implementation and validation flow from the start (Bhunias & Tehranipoor, 2019).

4.a What are the six steps involved in a practical fault injection attack? Explain each step with a focus on the "Bumping" technique to extract contents of the protected memory?

A practical fault injection attack generally follows a structured process, although the exact wording can vary depending on the target or attack method. The attack usually moves through six main steps: target selection and asset identification, reconnaissance and preparation, trigger point identification, fault injection or “bumping”, observation and extraction of faulty outputs and iterative exploitation (Maswood, 2026).

1. Target selection and asset identification

The attacker first chooses the hardware target and identifies what protected asset is worth attacking. In many cases this is protected memory, secret keys, firmware or authentication logic. This step is important because the attacker needs to know what information they want to extract and where it is likely stored.

2. Reconnaissance and preparation

Next, the attacker studies how the device operates. This may include learning the power behavior, clock behavior, boot process, memory layout and security checks. The attacker also prepares the tools needed for the attack, such as glitching equipment, probes, timing controls and other hardware used to inject the fault.

3. Locate the vulnerable point

The attacker identifies the exact moment where a security mechanism can be disrupted. Once a weak spot is found, the attacker can then perturb and exploit it. For memory extraction, this weak point is often during a protection check, authentication step or access-control decision related to the protected memory (Maswood, 2026).

4. Inject the fault using bumping technique

This is the actual attack step where the attacker applies a very brief and controlled disturbance to the target system during a critical operation. This bump may take the form of a voltage glitch, clock glitch, light exposure or another fault injection method. The main purpose is to knock the system out of its normal behavior for just long enough to cause an error. If successful, the device may skip a security check, misread a lock bit or improperly allow access to protected memory.

5. Observe the faulty behavior and extract data

After the bump is applied the attacker will examine how the device is behaving differently. If the injected fault worked, the target may reveal memory contents, output incorrect but useful data or even allow access to regions that were previously protected. At this stage, the attacker will begin to attempt to read and capture contents of protected memory.

6. Iterative exploitation

Fault injection attacks are rarely successful on the first try so the attacker usually repeats the process many times while making adjustments to timing, voltage level, clock offset or location of the bump until the desired faulty behavior is achieved consistently.

Once the right condition is found the process can then be repeated or automated to extract larger amounts of protected data.

The bumping technique is effective because it temporarily disrupts the system at the exact moment a memory protection mechanism should be enforced. Instead of permanently damaging the chip, the attack tries to create a short-lived fault that causes the hardware to make a security mistake. If successful, this can expose firmware, secret values and other protected contents that should not normally be readable (Maswood, 2026).

References

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- Paar, C., & Pelzl, J. (2014). *Understanding cryptography: A Textbook for Students and Practitioners*. Springer.